# Thomas Harriot and the Mercator Map

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This talk was written for a reunion of former students in Mathematics and Computer Science held at Oriel College on 30th November, 2024.

It is in large part based on Jon V. Pepper's article, 'Harriot's calculation of meridional parts', *Archive for History of the Exact Sciences*, 1968.

Suppose you wanted to fly from New York to London. The shortest way is to follow a great circle, the intersection of the earth's surface with a (mathematical) plane that contains New York, London, and the centre of the Earth. If, on reaching London, the pilot decided to continue on the same course, you would eventually come full circle and end up where you started.

As you sit in business class, you look at the map in the back of the in-flight magazine, and there the great circle route looks less sensible, because it curves well to the North of the apparent track between source and destination: in fact, towards the middle of the flight it visits latitudes that are further North than both endpoints. Though London is further from the equator than New York, in the last part of the flight the plane is steering towards the South side of East.

That leads to the question flight attendants must be asked so often: why don't we just fly straight there? The answer is that the apparent straight-line path is in fact longer than the great circle route – in this case, by about 4%. That straight path is a *rhumb line*, a course with a constant bearing throughout, and in the days before electronic navigation aids, would be much easier to steer than a great circle with a constantly changing bearing.

If the pilot follows the rhumb line and forgets to land in London, the puzzle arises what will happen to the plane subsequently.

# Continuing the rhumb line

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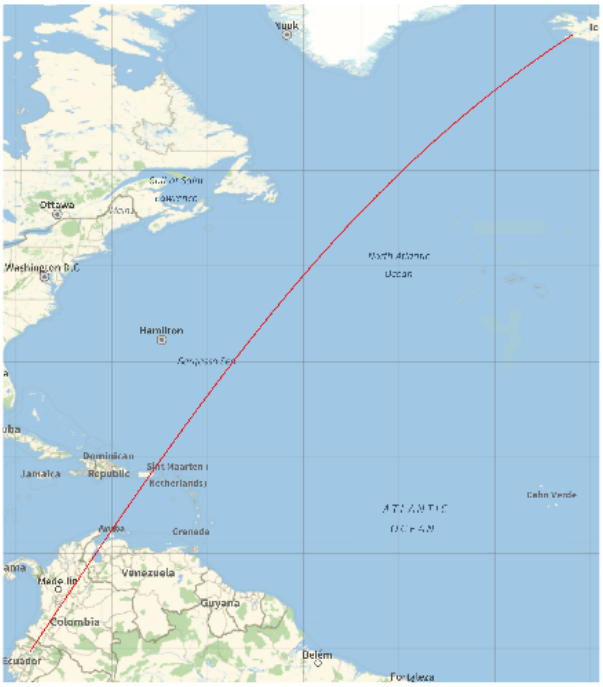
Since the course is always to the North of due East, the plane must get closer and closer to the North pole, and in fact it traces out a spiral course. The nature of this spiral will be important later when we look at the mathematical properies of rhumb lines.

# Making a map

On a *plane chart*, rhumb lines appear curved.

Worse still, measured bearings are wrong.

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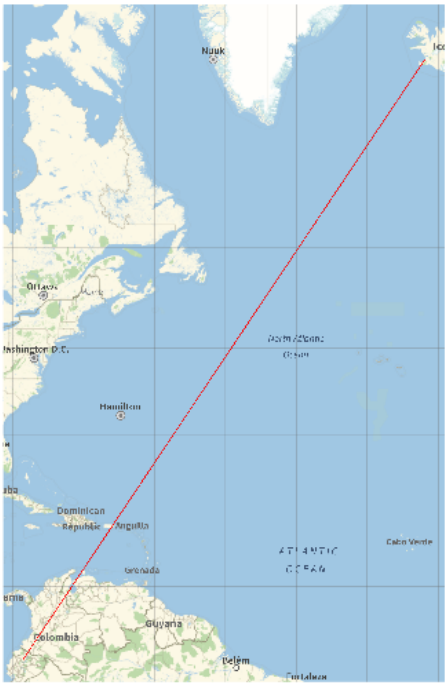
Crudely made charts, even if they plot latitude and longitude on a rectangular grid, suffer from the problem that rhumb lines do not appear accurately straight. Early *plane charts*, in popular use before 1550, plotted latitude and longitude on uniform scales. Over moderate distances, the discrepancy is small, but this map shows the rhumb line from Quito to Rekjavik on an equirectangular projection, and it is noticeably curved. What's worse, measuring the angle between the course and the meridian lines (or measuring the same angle for a stright line drawn on the map) doesn't give a correct course to steer.

# Gerardus Mercator (1512-94)

Huge world map of 1569, made from 18 sheets pasted together.

Stretched vertically at higher latitudes so as to make rhumb lines straight.

But how much? Mercator seems to have used empirical methods.

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This problem was solved, after a fashion, by Gerardus Mercator in the middle of the sixteenth century. He published a huge map of the known world, pasted together from 18 engraved and printed sheets. At over two metres wide, the map was far too large for use in navigation, but it made the vital innovation of stretching the map at higher latitudes so as to make the rhumb lines straight. Smaller extracts of the map with the same stretching of the vertical scale would be useful in practical navigation, making it possible to take bearings off a chart and steer a rhumb line course reliably. This was a vast improvement on earlier navigational methods such as *parallel sailing*, that is, sailing due North or South until on the correct latitude, then completing the journey by sailing along a parallel. This was the only viable method in an age before navigators had chronometers to help them determine their longitude: that would come two centuries later.

The map needs stretching, but by how much? Mercator seems to have used empirical methods, such as drawing rhumb lines on a physical globe using a metal angle template, then reading off latitude and longitude for plotting.

# Thomas Harriot (c.1560-1621)

- Undergraduate at St Mary's Hall 1577–80.
- Protégé of Walter Raleigh, and later Henry Percy, Earl of Northumberland.
- Mostly known for his work in algebra.
- Participant in early colony in North America.
- Left behind 1000's of pages of *largely unexplained* calculations.

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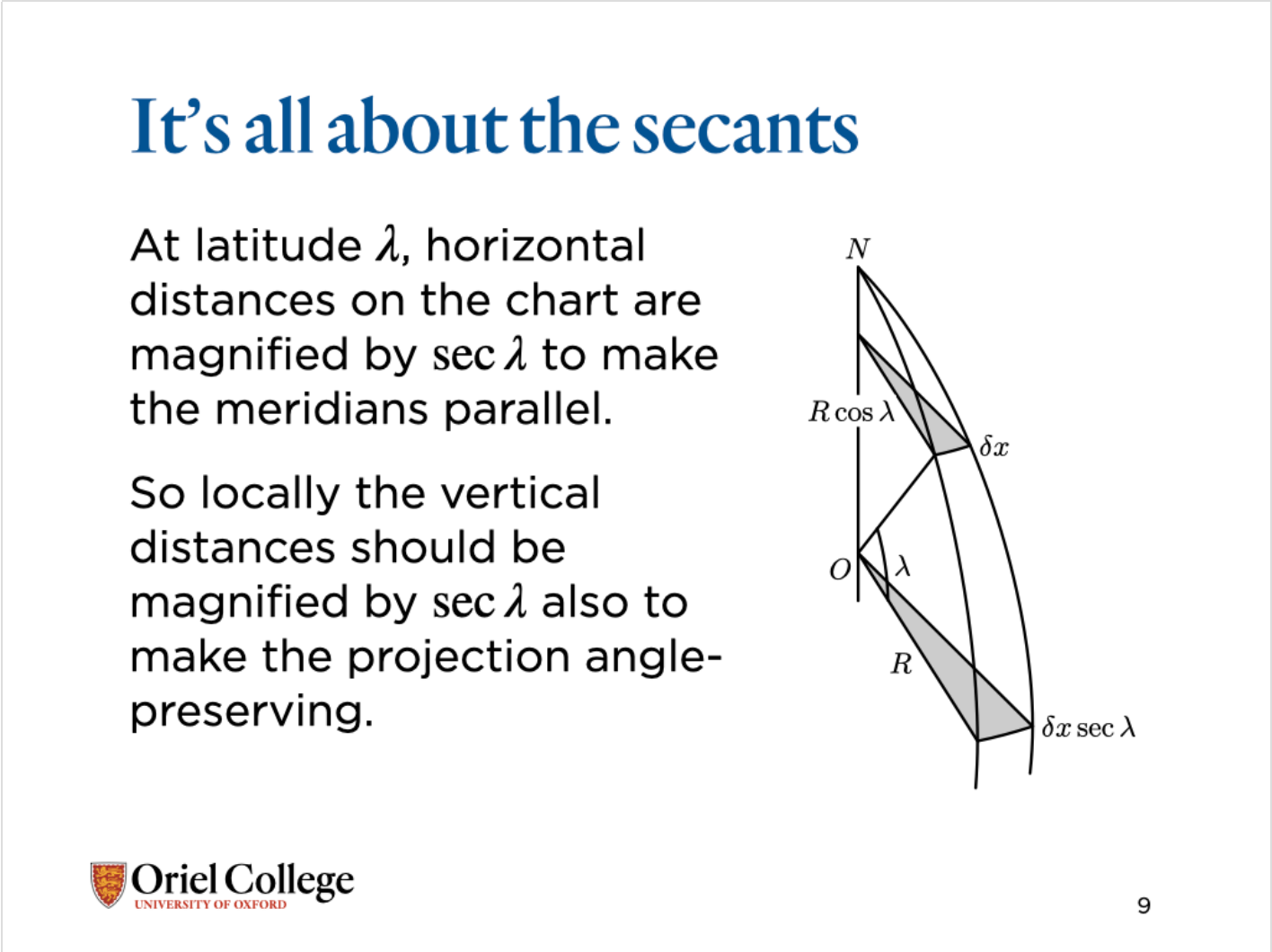
This is where our hero comes into the picture. Thomas Harriot was an Elizabethan mathematician who was born in Oxfordshire in about 1560, and studied at St Mary Hall in Oxford from 1577 to 1580. At the time, only well-off undergraduates could afford to be members of an actual college, and poorer students belonged instead to a private hall, with each college surrounded by a number of halls that depended on it. For example, the institution that is now the college St Edmund Hall was once a private hall in the orbit of Queens'. Almost all the private halls that once existed have either turned into colleges themselves or been absorbed into their parent college. A recent example is Mansfield College, which (though it has College in its name) was a provate hall until 1995.

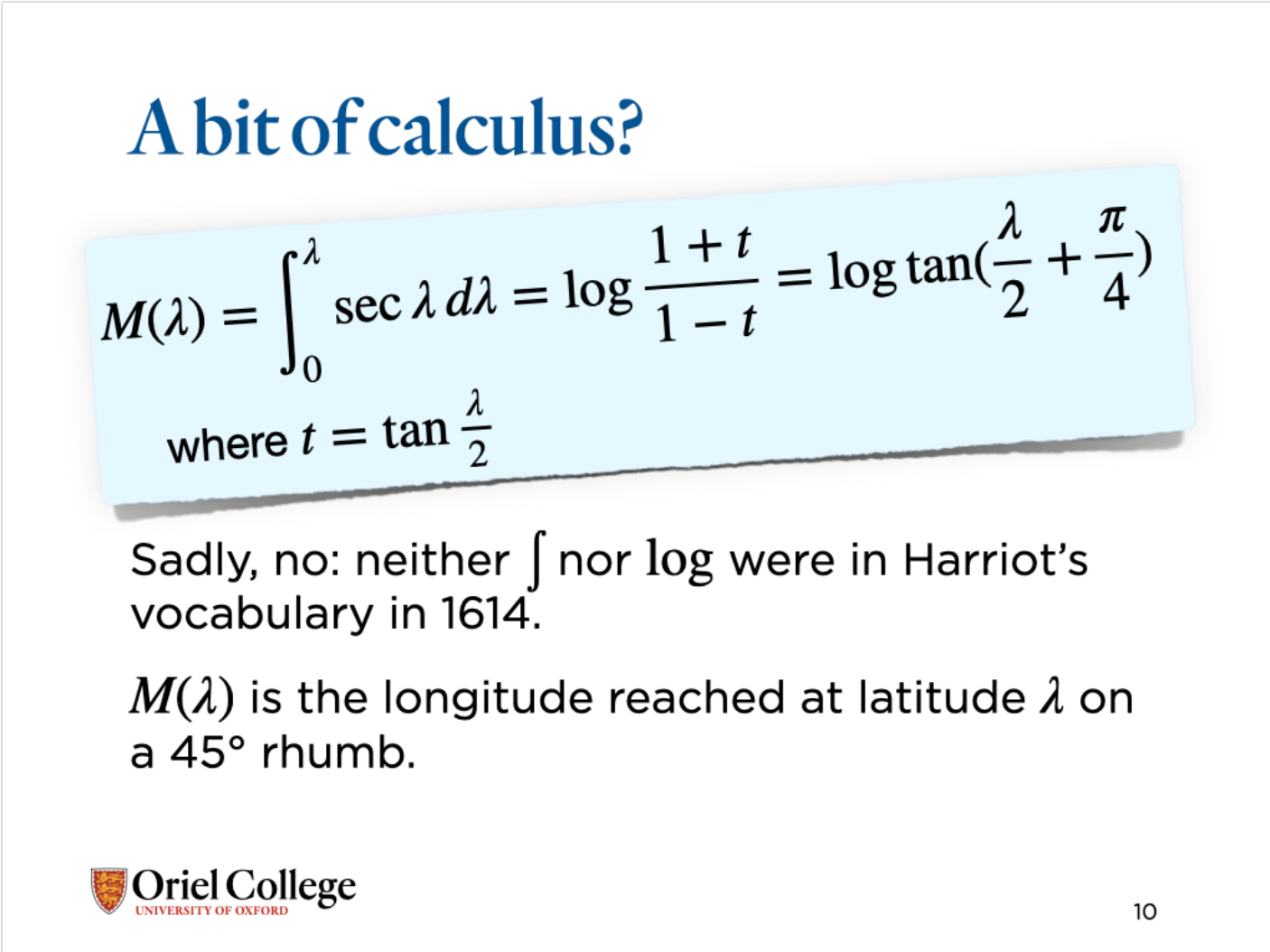
Anyway: St Mary Hall occupied what is now the third quad until it was absorbed into Oriel in 1902, and it was there that Harriot was an undergraduate. During his time in Oxford, he became associated with Walter Raleigh, who had matriculated at Oriel in 1572 but apparently never came into residence. On leaving Oxford, Harriot for a time ran a school of navigation for Raleigh in London, and he later became a participant in an early colony sponsored by Raleigh in North America, leading to his only publication, a sort of prospectus for colonial opportunities in America.

Later, Harriot would become a client of the 'wizard earl' of Northumberland, Henry Percy, and given a house and a pension and the freedom to pursue his own researches in astronomy, chemistry and mathematics. He is best known for his work in algebra, where he contributed greatly to methods for solving polynomial equations. Though he published little during his lifetime, he left behind thousands of pages of diverse calculations that have fascinated historians of mathematics on and off ever since. Recently, his entire manuscript legacy has been scanned, collated and made available online. Oh, incidentally, this picture in the collection of Trinity College, though Oriel has a recent copy, is almost certainly not a portait of Harriot.

What I want to describe in this talk is Harriot's solution to the problem of laying out a Mercator map, which he seems to have developed in 1594, completing the detailed calculations later, probably in 1614–15.



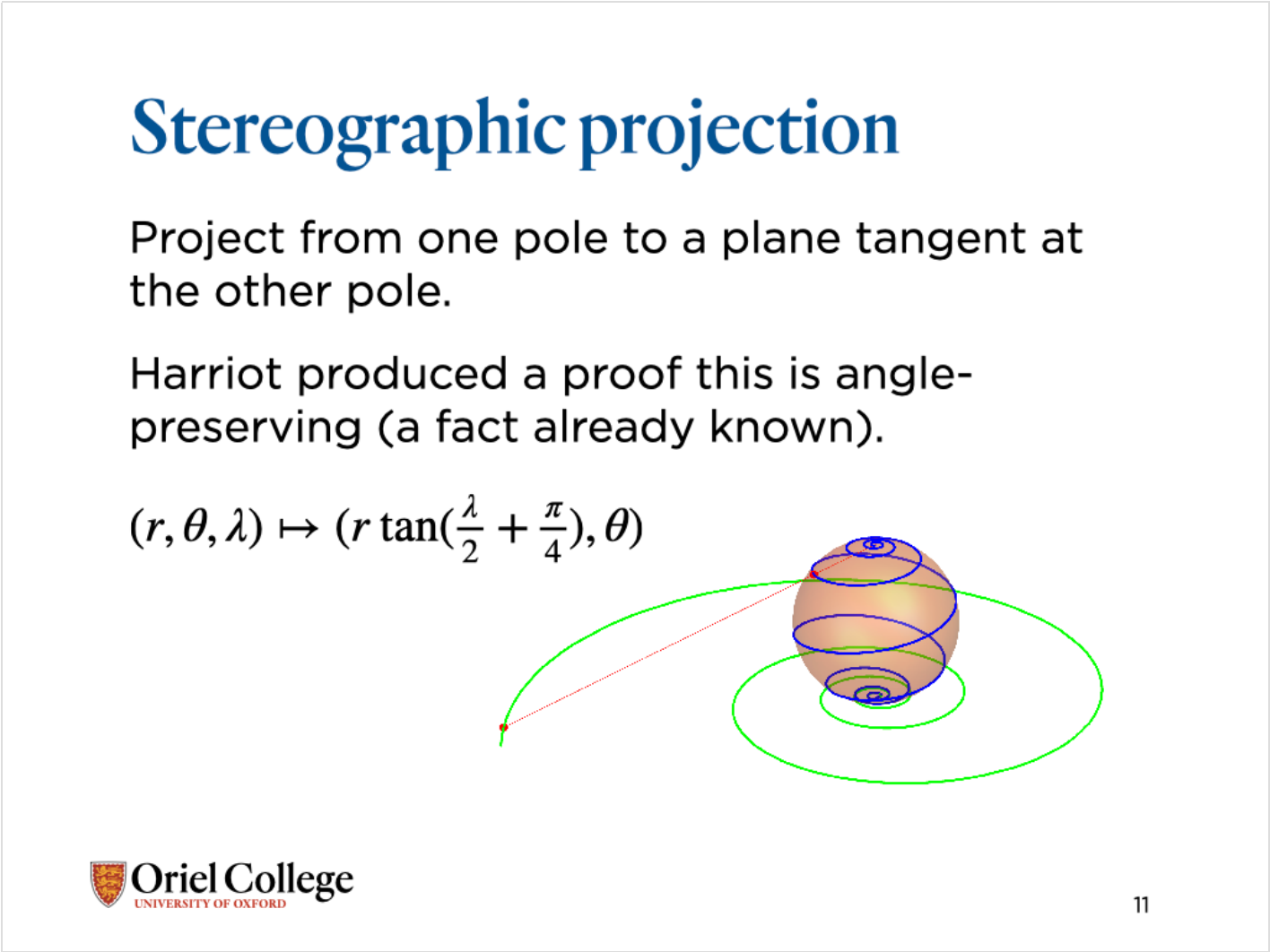

The problem is this: we know that the gap between lines of longitude must be stretched more and more as the latitude increases in order to make the meridians parallel on the map. The local horizontal scale factor is $\sec\lambda$, where $\lambda$ is the latitude; so to make the projection angle-preserving and shape-preserving (*conformal*), we should make the local scale factor be $\sec\lambda$ vertically too. That much was known, and earlier workers such as Edward Wright (1561–1615) had derived an approximate vertical scale by adding up secants from minute to minute of arc, starting at the equator. This did, to be fair, give results that were adequate for practical purposes at moderate latitudes.




Why not just use calculus, and integrate the secant function to give $M(\lambda)$, the correct vertical position of latitude $\lambda$? This is a standard calculus problem, inflicted on generations of the brighter A level students by maths teachers wanting a few moments peace. Perhaps some members of the audience have had the problem inflicted on them – and perhaps some have become maths teachers themselves and inflicted it on others.

It can be solved by everyone's favouite substitution, $t = \tan(x/2)$, or here $\tan(\lambda/2)$, and the answer can be expressed in various forms, among them $\log\tan(\lambda/2 + \pi/4)$, a form that I like because we will recognise parts of it later.

But all this is moot, because in Harriot's time the idea of integration had not been invented, and neither had the $\log$ function, some properties of which would be discoved by John Napier in the years that followed Harriot's investigations. So if Harriot was to solve the problem of finding the *meridional part* $M(\lambda)$ for any given latitude $\lambda$, he would have to go about it by a different route. It's tempting to view Wright's approach of adding up secants minute by minute as a form of numerical integration, and mathematically that's correct. But it isn't accurate historically, because Wright didn't have the concept of integral and so couldn't see it that way. He and Harriot knew that it was an approximation, but didn't have the machinery to know how accurate it would be. By way of contrast, Harriot's method would yield results as accurate as could be desired, and would do so for independently for each latitude.




Harriot's solution begins with a different projection from Mercator's, one that is genuinely a projection geometrically. It is the *stereographic projection*, obtained by placing a light at one of the poles and casting shadows on a plane tangent to the other pole.

This projection is conformal, that is, it preserved angles, in the sense that the angle between two lines in the tangent plane at any point on the sphere is that same as the angle in the projection plane between the images of those lines. This was a fact known in the ancient world and exploited in the design of astrolabes, but first proved by Harriot, though like most if his work the proof was never published.

The projection maps the spiral of a rhumb line (a *loxodrome*) into a spiral in the plane. In the picture. it's easy to see that part of the spiral in the Southern hemisphere maps to a spiral in the part of the plane nearest the South pole. Harder to see, but still true, is that the spiral in the Northern hemisphere maps to a path in the plane that continues to spiral outward even as the loxodrome approaches the North pole.

The point $(r, \theta, \lambda)$ – where $\theta$ is the longitude and $\lambda$ the latitude – is mapped to a point in the plane with radius $r \tan(\lambda/2 + \pi/4)$. [I dropped a factor of 2 here.] In this formula, there's a quantity $\pi/2$ that comes from converting between latitude $\lambda$ and polar angle $\phi$, and a factor of $1/2$ that comes from the relationship between the angle subtended by an arc at the centre of the sphere and at the pole. So there's a geometric argument that accounts for these elements of the formula that we can now derive by calculus.

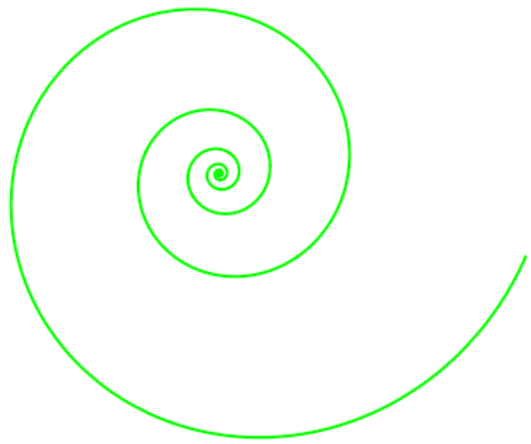
Now let's consider the spiral in the plane that is the image of a loxodrome on the globe.

# Equiangular spiral

The image of a rhumb line is an equiangular (or "logarithmic" spiral).

The radius increases by a constant ratio as it sweeps out equal angles.

$$r_n/r_o = (r_1/r_0)^n$$



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Because the stereographic projection preserves angles, and because the loxodrome makes a constant angle with the meridians, the image spiral makes a constant angle with radius vectors. It is an equiangular spiral (or what we could anachronistically call a logarithmic spiral).

If we consider a series of radius vectors that are spaced at equal angles, we can see that each segment of the spiral is similar to all the others, and the radius itself increases by a constant factor in each segment. So we can conclude that each radius $r_n$ is a constant $r_0$ times the $n$'th power of a fixed ratio.

# The fundamental equation

Let $\beta < 1$ be the ratio $r_1/r_0$ for a 45° spiral and an angle of (say) $\theta = 1'$. If

$$\tan(\frac{\pi}{4} - \frac{\lambda}{2}) = \beta^n$$

then $M(\lambda) = n\theta$. [Actually $\beta = e^{-\theta}$.]

To find $M(\lambda)$ for any latitude $\lambda$, determine $n$ such that

$$\beta^{n+1} < \tan(\frac{\pi}{4} - \frac{\lambda}{2}) \leq \beta^n$$

then $n\theta \leq M(\lambda) < (n+1)\theta$; use linear interpolation.

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Pick $\theta$ to be a unit of angle, say one minute, and find the associated ratio $\beta = r_1/r_0$. Harriot liked to work with numbers less that one, so picked a spiral that moved inwards with increasing angle. It's easy to find the latitude whose mer-part is $n\theta$: it is that latitude $\lambda$ such that $\tan(\pi/4 - \lambda/2) = \beta^n$. The minus sign comes from the fact that we're following a course that spirals inward, or that the centre of projection is the South pole rather the North, or that we're measuring the latitude with the opposite sign convention.

Harriot needed to reverse this process: given $\lambda$, he wanted to find $M(\lambda)$. For this, we can find a value of $n$ such that the tangent is close to $\beta^n$, and then say $M(\lambda) \approx n\theta$, and for better accuracy we can use linear interpolation between adjacent values of $n$. This is the same process that is involved in the methods that John Napier and Henry Briggs used to compute logarithms. But note that linear interpolation between these points marked by rational powers of $\beta$ doesn't entail that irrational powers exist, though it does suggest it.

# A refinement

Find $\beta$ as above, then $\alpha = \beta^{60}$ and $\gamma = \beta^{1/100}$.

Make tables of powers up to $\alpha^{90}$ and beyond, $\beta^{60}$ and $\gamma^{100}$.

Now if

$$\tan(\frac{\pi}{4} - \frac{\lambda}{2}) \approx \alpha^a \beta^b \gamma^c$$

then $M(\lambda) \approx a^\circ\,(b + c/100)'$ gives degrees and minutes to two decimal places.

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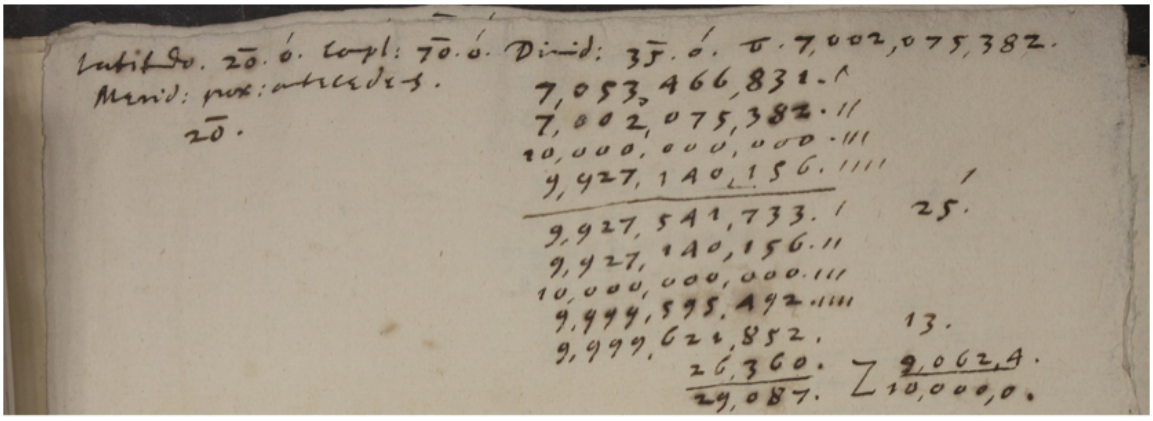
Harriot used a slightly refined method that involved two other quantities $\alpha$ and $\gamma$, with $\alpha = \beta^{60}$ corresponding to a mer-part of one degree, and $\gamma = \beta^{1/100}$ to a mer-part of a hundredth of a minute. For convenience, he prepared tables of powers of $\alpha$, $\beta$ and $\gamma$. Now expressing the tangent in the form $\alpha^a \beta^b \gamma^c$ gives a mer-part in degrees, minutes, and hundredth parts of a minute.

Expressing lengths as angles perhaps needs an explanation or a reminder: one degree corresponds to the distance between adjacent parallels of latitude on the globe, or between meridians at the equator. One minute of arc at the equator is equal to one nautical mile, so even without interpolation, Harriot obtained answers accurate to the hundredth part of a mile.

# Calculating $M(20°)$




$M(20°) = 20° 25'.13\,90624$

Dozens of pages with such calculations exist!

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The manuscripts contain dozens of pages that record calculations like this one, supported by division sums that are not part of the manuscript record. This calculation finds the mer-part of latitude $20°$. At the top, you can see *Latitudo 20.0'*, and also its complement, 70.0', and half that, 35.0' and its tangent, [0.]7,002,075,382. This corresponds to the quanity $\tan(\pi/4 - \lambda/2)$. There follows a calculation on several lines, from which we can pick out the figures 20, 15, and 13, making up the result $20° 25'.13$. There's a large Z at the bottom that denotes values in proportion.

In some respects, Harriot's notation is superior to what we use: it's nice that he has a special symbol – a circle with a tangent line above it – for the tangent function. On the other hand, the lack of the decimal point makes everything much more difficult to follow. It was John Napier who introduced, or at least popularised, the use of decimal points, and Harriot worked explicitly with fractions whose denominators were powers of ten – hence the quantity 10,000,000,000 shown here.

Let's transcribe the rest of the calculation into modern notation to see what's going on.

# In modern notation

Latitude 20°; complement 70°; halved 35°

| | | |
|---|---|---|
| tan 35° | = | 0.7002075382 |
| $\alpha^{20}$ | = | 0.7053446831 |
| quotient | = | 0.9927140156 |
| $\beta^{25}$ | = | 0.9927541733 |
| quotient | = | 0.9999595492 |
| $\gamma^{13}$ | = | 0.9999621852 |
| difference | = | 26360 |
| $\gamma^{13}-\gamma^{14}$ | = | 29087 |
| quotient | = | 0.90624 |

$M(20°)$ = 20° 25'.13 90624

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Given the tangent 0.7002..., we first find the last power of $\alpha$ from the table that is still larger than the tangent: it is $\alpha^{20} = 0.7053\ldots$. Then compute the quotient $\tan/\alpha^{20}$, and find the last power of $\beta$ that is larger, namely $\beta^{25}$. Again the quotient is calculated, and the process is repeated with $\gamma^{13}$. The last part of the calculation compares the difference $\gamma^{13} - \tan/\alpha^{20}\beta^{25}$ and compares it with the tabulated value of $\gamma^{13} - \gamma^{14}$ to derive additional decimal places by linear interpolation.

# Let's check that!

$M(20^\circ)$ calculated as $20^\circ\,25'.13\,90624$

$$\begin{aligned} -\log\tan 35^\circ &= 0.3563785047 \\ &= 20.418984230^\circ \\ &= 20^\circ\,25'.13\,90538 \end{aligned}$$

The small error is largely explained by the approximate value Harriot used for $\beta$.

Harriot's table of meridional parts remained unsurpassed for 350 years.

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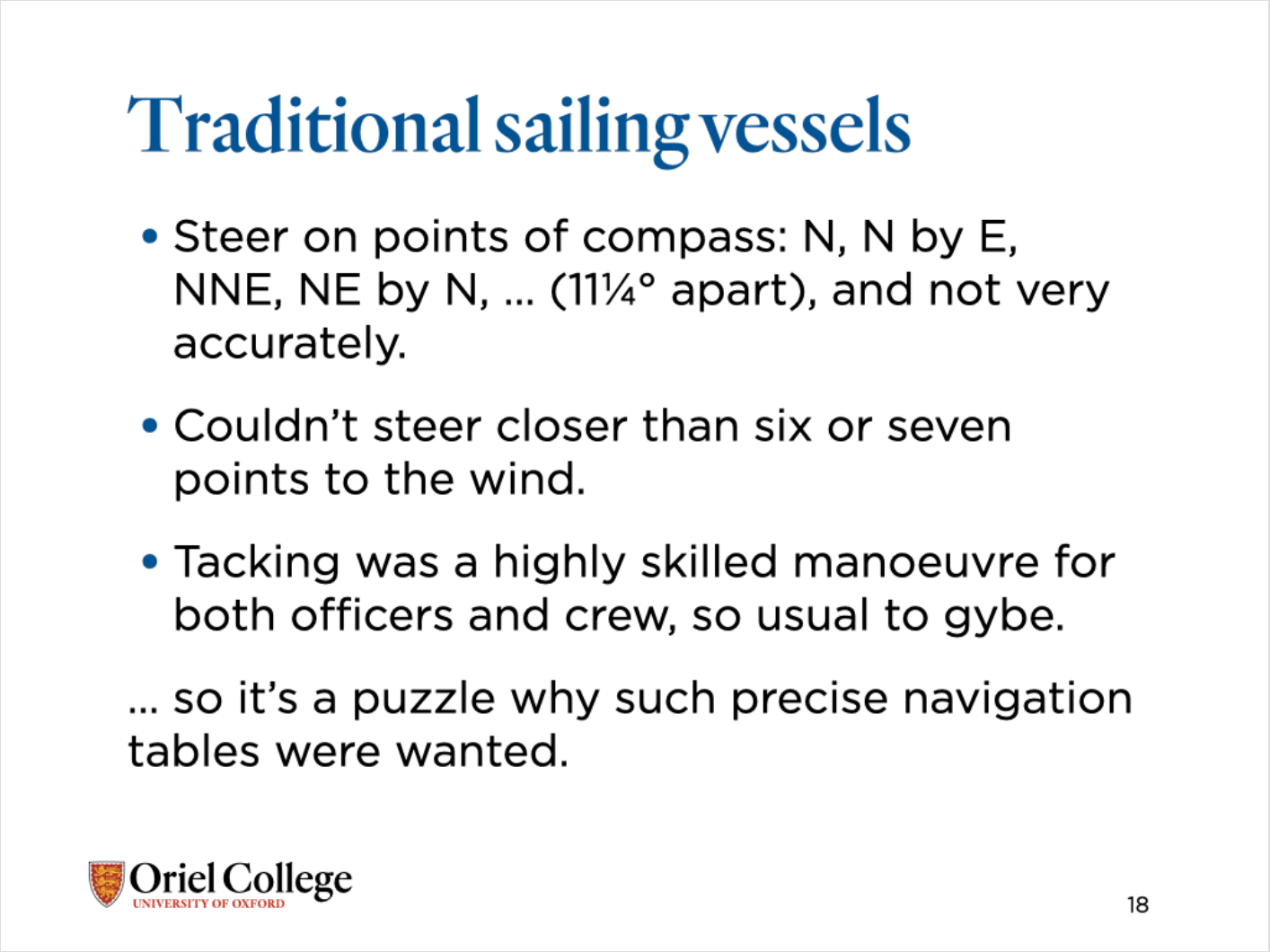
We can check on Harriot's result by using the formula derived by calculus, and modern tables or calculating devices, and we find that Harriot's result is accurate to four or almost five decimal places of minutes. The small error is largely explained by a slightly inaccurate value for $\beta$ that Harriot found with a geometrical approximation. Even tiny errors in the value of $\beta$ are magnified when it is raised to huge powers, as here.

Despite this inaccuracy in the parameters, Harriot's method gives meridional parts with no further steps of approximation. The values he computed remained unsurpassed in accuracy for more than 350 years.



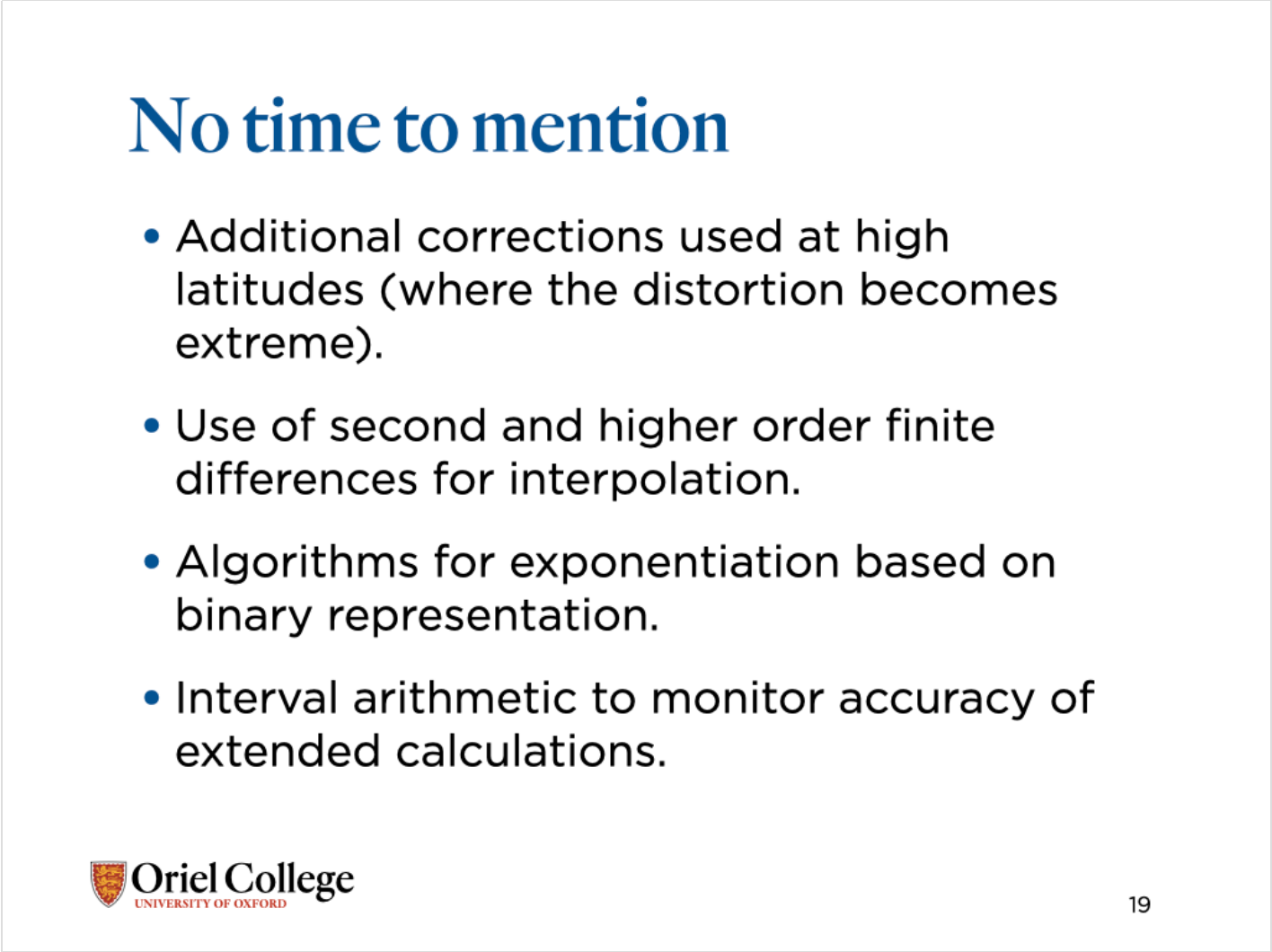

What remains a puzzle for me is why Harriot found it desirable to compute these values so accurately, in a world where both the performance of sailing ships and standards to navigation where very crude.

A typical square-rigged vessel would be equipped with a magnetic compass, but awareness of magnetic variation, the difference between North as shown on the compass and the true North of the meridian, was only just developing. It was traditional to steer a vessel on one of the points of the compass that were spaced 11-1/4 degrees apart, and pointless to attempt anything more precxise, given the limited accuracy of the compass, the unknown magnetic variation, and the huge leeway of the vessel.

Modern dinghys and leisure yacht with their fore-and-aft sails can make progress upwind, usually steering to within 45 degrees of the facing wind ("four points off the wind"), and tacking to make progress along the desired course. In a traditional square-rigged vessel, things were not so easy. It was impossible to steer closer than six or seven points to the wind, and sailing performance was in any case markedly better on a broad reach than close-hauled.

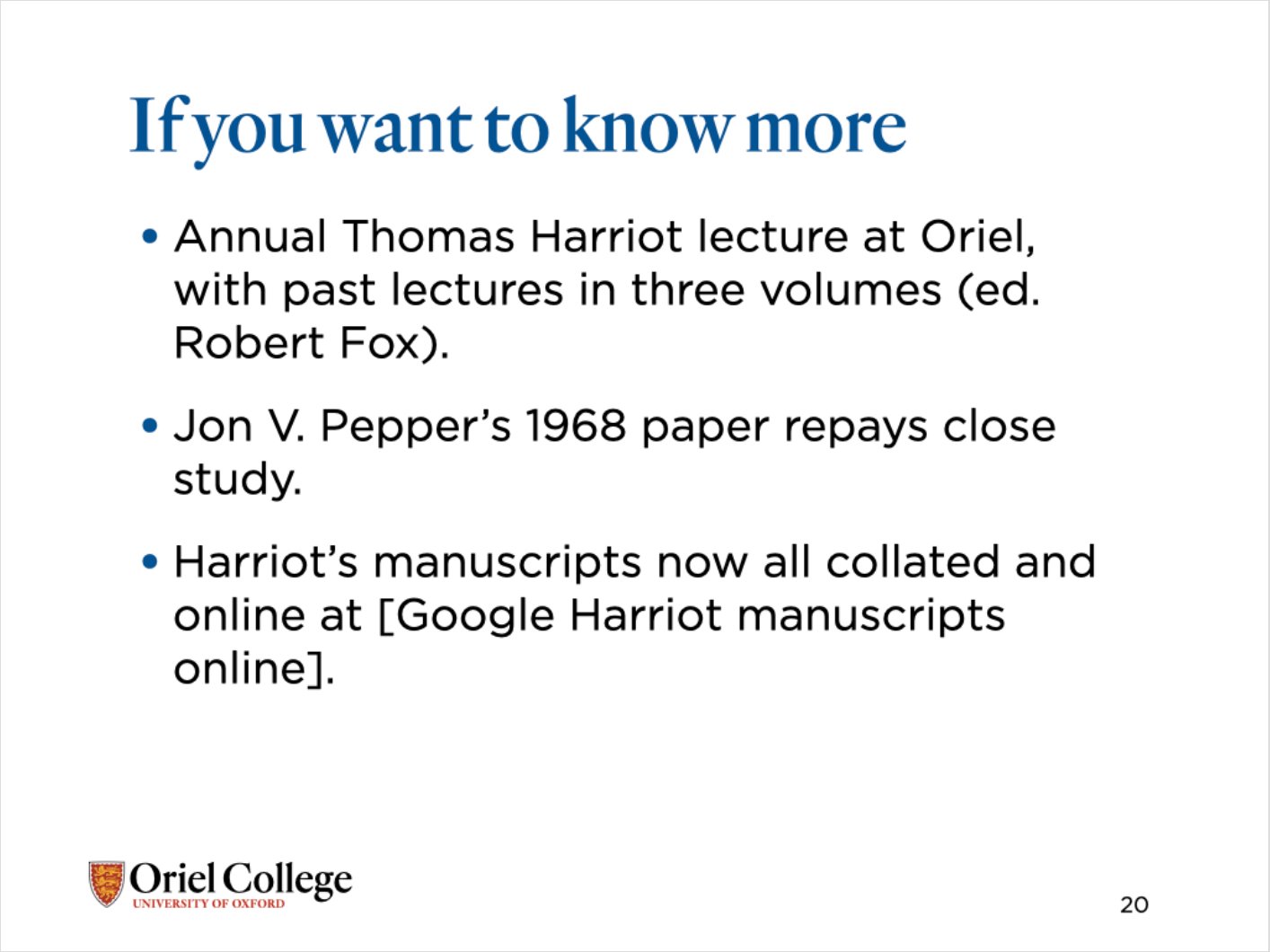
As for tacking, coming round from so far off the wind was a great challenge of timing for both the officer and a highly skilled crew, so it was more usual in ordinary circumstances to go round the other way, and gybe in order to change tack. All this meant that for transatlantic sailing, the sensible thing was to head south in order to pick up the trade winds, and not to attempt rhumb line sailing at all.

That's why it remains a puzzle to work out what motivated Harriot's painstaking work, unless it was simply the pleasure of calculating to the maximum achievable accuracy.




There's no time in this talk to do any more than mention other aspects of Harriot's mathematics that impinge on his navigational work, still less those aspects that don't impinge on it. The account of his calculations given earlier doesn't cover the additional steps needed at high latitudes, where the Mercator stretching became more extreme, but also the entries in the tangent tables (close to zero degrees) would start to lose significant figures.

As well as linear interpolation, Harriot was aware of and used second and higher-order methods that in some ways anticipated Newton's interpolation formulas – though he didn't always get the coefficients right. The navigational work and some other work on compound interest involved raising numbers (here slightly less than unity) to immense powers, and Harriot used several modern-looking ideas: explicit use of a binary representation of the exponent in order to maximise the use of squaring to reach high powers (in an algorithm still used for cryptography); the use of scientific notation to deal with huge numbers, despite the lack of decimal points; and finally, the use of interval arithmetic to get upper and lower bounds on the true result of a calculation.




If you want to know more about Harriot, a good place to start is the series of annual Thomas Harriot lectures held at Oriel for many years. Three volumes of past lectures have been published in book form under the editorship of Robert Fox, and can be found in the college library. Not all the lectures are directly about Harriot himself, but sometimes about his cultural context, and relatively few of them directly address his mathematics, historians by their nature being more interested in wit and poetry. But nevertheless a lot of the atmosphere of Harriot's orbit in Elizabethan England comes through.

More specifically, the best work on Harriot's contributions to navigation were undoubtedly made by Jon V. Pepper in the 1960's, and the paper on which I've based this talk repays very close study. Besides the fact that one idle afternoon Harriot invented binary arithmetic, there's a careful focus on methods of computing (indeed, on algorithms) throughout Harriot's work that makes me want to claim his as Oriel's first Computer Scientist. It would be good to say that Computer Science has been studied here since the 16th century.

Finally, a boundless source of delight is Harriot's manuscripts, now all scanned, collated, and freely available online. Previously, the manuscript pages were divided between the British Museum and the Duke of Northumberland's residence at Sion House, and not in any logical way. Now they are all viewable from anywhere in the world, and organised into a proper order. If you have ever filled a page with numbers just for the joy of calculating, then you will find a kindred spirit in Harriot.

- Slides

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